

2016.0 RANGE ROVER (LG), 206-00

BRAKE SYSTEM – GENERAL INFORMATION

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the brake system, refer to the relevant description and operation sections in the workshop manual.

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

Visually examine the front and rear wheel and tire assemblies for damage such as uneven wear patterns, tread worn out or sidewall damage. Verify the tires are the same size, type and, where possible, same manufacturer.

Replace the damaged wheel or excessively worn tire

Wheels and tires must be cleared of any foreign matter and tire pressures adjusted to the correct specification

If the tires exhibit uneven wear or feathering, the cause must be corrected. Check the steering and suspension components for damage or wear and, if necessary, check and adjust front wheel alignment

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Brake master cylinder ▪ Brake caliper piston(s) ▪ Brake discs ▪ Wheel bearings ▪ Brake pads ▪ Power brake booster ▪ Brake pedal linkage ▪ Brake booster vacuum hose ▪ Tires ▪ Debris	▪ Electric park brake actuator ▪ Electric park brake control module ▪ Electric park brake switch ▪ Damaged or corroded wiring harness ▪ Brake master cylinder fluid level switch

Road Test

Carry out a road test to compare actual vehicle braking performance with the performance standards expected by the driver. The ability of the test driver to make valid comparisons and detect performance deficiencies will depend on experience

The driver should have a thorough knowledge of brake system operation and accepted general performance guidelines to make good comparisons and detect performance concerns

An experienced brake technician will always establish a route that will be used for all brake diagnosis road tests. The roads selected will be reasonably smooth and level. Gravel or bumpy roads are not suitable because the surface does not allow the tires to grip the road equally. Crowned roads should be avoided because of the large amount of weight shifted to the low set of wheels on this type of road. Once the route is

established and consistently used, the road surface variable can be eliminated from the test results

Before a road test, obtain a complete description of the customer concerns or suspected condition. From the description, the technician's experience will allow the technician to match possible causes with symptoms. Certain components will be tagged as possible suspects while others will be eliminated by the evidence. More importantly, the customer description can reveal unsafe conditions which should be checked or corrected before the road test. The description will also help form the basic approach to the road test by narrowing the concern to specific components, vehicle speed or conditions

Begin the road test with a general brake performance check. Keeping the description of the concern in mind, test the brakes at different vehicle speeds using both light and heavy pedal pressure. To determine if the concern is in the front or rear braking system, use the brake pedal and then use the park brake control. If the condition (pull, vibration, pulsation) occurs only with the park brake, the concern is in the rear brake system

If the concern becomes evident during this check, verify it fits the description given before the road test. If the concern is not evident, attempt to duplicate the condition using the information from the description

If a concern exists, use the symptom chart in order to isolate it to a specific sub-system and condition description. From this description, a list of possible sources can be used to further narrow the cause to a specific component or condition

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSE	ACTION
Brakes noisy	▪ Brake pads ▪ Brake discs	GO to Pinpoint Test A.

Vibration when brakes are applied	▪ Wheels/tires out of balance ▪ Wheel hub nuts loose ▪ Brake caliper mounting bolts loose ▪ Brake pads ▪ Foreign material/scratches/corrosion on brake disc contact surfaces ▪ Excessive brake disc thickness variation ▪ Excessive brake disc runout ▪ Wheel bearing wear or failure ▪ Suspension bushing wear or failure ▪ Steering bushing wear or failure	▪ GO to Pinpoint Test B. ▪ For a detailed description refer to the brake disc and wheel hub run out check. REFER to: (204-00 Suspension System - General Information) Front Brake Disc and Wheel Hub Runout Check - With Wheel On (General Procedures), Rear Brake Disc and Wheel Hub Runout Check - With Wheel On (General Procedures).
The brakes pull or drift	▪ Tire pressures/wear ▪ Brake calipers ▪ Brake pads ▪ Brake discs ▪ Wheel alignment adjustment ▪ Wheel bearing ▪ Suspension bushings and ball joints	GO to Pinpoint Test C .
The pedal feels spongy	▪ Air in brake hydraulic system ▪ Leak in hydraulic system ▪ Brake booster/master cylinder ▪ Brake pads	GO to Pinpoint Test D .
The pedal goes down fast	▪ Air in brake hydraulic system ▪ Leak in hydraulic system	GO to Pinpoint Test E .

	▪ Brake booster/master cylinder ▪ Brake pads	
The pedal goes down slowly	▪ Air in brake hydraulic system ▪ Brake booster/master cylinder	GO to Pinpoint Test F.
Excessive brake pedal effort required	▪ Brake pads ▪ Brake booster	GO to Pinpoint Test G.
Brake lockup during light brake pedal force	▪ Brake pads ▪ Brake calipers	GO to Pinpoint Test H.
Brakes drag	▪ Park brake applied/malfunction ▪ Seized park brake cables ▪ Seized brake caliper slide pins ▪ Seized brake caliper ▪ Brake booster ▪ Pedal gear	GO to Pinpoint Test I.
Excessive/Erratic brake pedal travel	▪ Hydraulic system ▪ Brake pads ▪ Brake discs ▪ Hub and bearing assembly	GO to Pinpoint Test J.
The red brake warning indicator is always on	▪ Fluid level ▪ Brake master cylinder fluid level sensor ▪ Park brake control ▪ Electrical circuit	Fill the system to specification. Check for leaks. Install a new brake master cylinder fluid reservoir as required. Refer to the relevant section in the workshop manual for electric park brake control module and circuit tests
Slow or incomplete brake pedal	▪ Brake pedal binding ▪ Brake booster/master	GO to Pinpoint Test K.

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PINPOINT TESTS

PINPOINT TEST A : BRAKES NOISY

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: INSPECT BRAKE PADS	
	1 Inspect the condition of the front and rear brake pads. Check for damage to any anti-squeal shims
	Are the brake pads OK? Yes GO to A2. No Clean/install new front and rear brake pads as required. Retest vehicle for brake noise
A2: INSPECT BRAKE DISCS	
	1 Inspect the brake discs for excessive corrosion, wear or disc thickness variation
	Does excessive corrosion, wear or disc thickness variation exist? Yes Install new front and rear brake discs and brake pads as required. Retest vehicle for brake noise No No action required, vehicle is OK

PINPOINT TEST B : VIBRATION WHEN BRAKES ARE APPLIED

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: ROAD TEST VEHICLE	
	1 Road test the vehicle between 40-80 km/h (25-50 mph) without applying brakes
	Is the vibration present? Yes Refer to the relevant section in the workshop manual for noise vibration and harshness tests No GO to B2.

B2: CHECK FOR BRAKE VIBRATION

	1 Road test the vehicle between 40-80 km/h (25-50 mph) with light and medium application on the brake pedal
	Is a vibration present? Yes Check the brake caliper mounting bolts and wheel hub nuts and tighten to specification as required. Check the balance of all road wheels and tires and repair as required. Check the brake discs for excessive wear, runout, thickness variation or cracks. Install new brake discs and brake pads as required GO to B3. No No action required, vehicle is OK

B3: IS VIBRATION STILL PRESENT UNDER BRAKE APPLICATION?

	1 Road test the vehicle between 40-80 km/h (25-50 mph) with light and medium application on the brake pedal
	Is a vibration present? Yes Check for wear or failure of steering gear bushings. Check for wear or failure of steering gear ball joints. Check for wear or failure of front wheel bearings, suspension bushings and ball joints. Check for wear or failure of rear wheel bearings, suspension bushings and ball joints. Refer to relevant section in workshop manual and install new components as required No No action required, vehicle is OK

PINPOINT TEST C : THE BRAKES PULL OR DRIFT

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: ROAD TEST VEHICLE	
	1 Road test the vehicle and apply the brake pedal
	Does the vehicle pull or drift? Yes GO to C2. No No action required, vehicle is OK
C2: INSPECT TIRE CONDITION/PRESSURE	
	1 Check for excessive tire wear or incorrect pressures
	Are the tires at the correct pressure and in good condition? Yes GO to C3. No

	Adjust the tire pressures or install new tires if excessively worn. Retest the system for normal operation
C3: CHECK CALIPERS	
	1 Check the disc brake caliper pistons and pins for binding, leaking or sticking
	Do the disc brake caliper pistons and pins bind, leak or stick? Yes Rectify sticking pins and install new brake calipers as required. Retest the system for normal operation No GO to C4.
C4: INSPECT BRAKE DISCS	
	1 Check the brake discs for excessive damage, thickness variation or runout
	Does excessive damage or runout exist? Yes Install new brake discs and brake pads as required. Retest the system for normal operation No GO to C5.
C5: INSPECT THE FRONT HUB AND WHEEL BEARING ASSEMBLY	
	1 Check the front hub and wheel bearing assembly
	Are the wheel bearings OK? Yes GO to C6. No Install new wheel bearings as required. Retest the system for normal operation
C6: CHECK SUSPENSION BUSHINGS AND BALL JOINTS.	
	1 Check all suspension bushings and ball joints
	Are the suspension bushings and ball joints OK? Yes GO to C7. No Install new front suspension bushings and ball joints as required. Install new rear suspension bushings and ball joints as required. Refer to the relevant section in the workshop manual
C7: CHECK VEHICLE ALIGNMENT	
	1 Check the vehicle alignment
	Is the alignment within specification? Yes

	No action is required, vehicle is OK No Adjust the alignment as required
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PINPOINT TEST D : THE PEDAL FEELS SPONGY	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
D1: CHECK FOR SPONGY PEDAL (ENGINE OFF)	
	1 Check for a firm brake pedal
	Is the brake pedal effort and brake pedal travel normal? Yes No action is required, vehicle is OK No GO to D2.
D2: CHECK BRAKE PEDAL RESERVE (ENGINE OFF)	
	1 Pump the brake pedal 10 times and hold on the final application
	Does the brake pedal feel firm on final application? Yes GO to D3. No Bleed the brake system
D3: CHECK BRAKE PEDAL RESERVE (ENGINE ON)	
	1 With engine running at idle speed
	2 Apply the brake pedal lightly three or four times
	3 Wait 15 seconds for the vacuum to recover
	4 Push down on the brake pedal until it stops moving downward or an increased resistance to the brake pedal travel occurs
	5 Hold the brake pedal in the applied position while increasing the engine speed to 2000 revs/min
	6 Release the accelerator pedal
	Does the brake pedal move downward as the engine speed returns to idle? Yes GO to D4. No Check the vacuum to brake booster
D4: CHECK BRAKE FLUID LEVEL	

	1 Check the brake master cylinder reservoir fluid level
	Is the fluid level OK? Yes Bleed the brake system. Retest the system for normal operation No Check for leaking brake system and rectify as required. Add fluid and bleed the brake system. Retest the system for normal operation

PINPOINT TEST E : THE PEDAL GOES DOWN FAST

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
E1: ROAD TEST VEHICLE	
	1 Road test the vehicle and apply the brake pedal
	Is the brake pedal effort and brake pedal travel normal? Yes No action required, vehicle is OK No GO to E2.

E2: CHECK BRAKE PEDAL TRAVEL-PRESSURIZE SYSTEM

	1 Pump the brake pedal rapidly (five times)
	Does the brake pedal travel build up and then hold? Yes Bleed the brake system. Retest the system for normal operation No GO to E3.

E3: CHECK FOR BRAKE SYSTEM LEAKS

	1 Check for external brake system leaks. For additional information, refer to brake master cylinder component test in this section
	Is there a leak present? Yes Repair as necessary, add fluid and bleed brake system. Retest the system for normal operation No No action required, system is OK

PINPOINT TEST F : THE PEDAL GOES DOWN SLOWLY

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
F1: ROAD TEST VEHICLE - CHECK BRAKE PEDAL OPERATION	

	1 Check if the condition occurs during actual stopping application by applying the brake pedal while the vehicle is moving
	Does the condition occur when the vehicle is moving? Yes GO to F2. No GO to F3.

F2: CHECK FOR BRAKE SYSTEM LEAKS

	1 Check for external brake system leaks. For additional information, refer to brake master cylinder component test in this section
	Are there any external brake system leaks? Yes Rectify as necessary. Add fluid and bleed the brake system. Retest the system for normal operation No GO to F3.

F3: CARRY OUT A BRAKE MASTER CYLINDER BYPASS TEST

	1 Test for brake master cylinder bypass condition. Refer to Brake master cylinder component test in this section
	Has a concern been identified? Yes Install a new brake master cylinder, add fluid and bleed the brake system. Retest the system for normal operation No No action required, system is OK

PINPOINT TEST G : EXCESSIVE BRAKE PEDAL EFFORT

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
G1: CHECK BRAKE PADS	
	1 Check the brake pads for wear, contamination, correct installation, damage and type
	Has a concern been identified? Yes Correctly install or install new brake pads as required. Retest the system for normal operation No GO to G2.
G2: CHECK VACUUM	
	1 Disconnect the vacuum hose from the brake booster

	2 Connect a vacuum/pressure tester to the vacuum hose
	3 Run the engine at normal operating temperature
	4 Record the vacuum reading
	Is the reading 40.5 kPa (12 in-Hg) or greater? Yes GO to G3. No Locate and rectify the source of low vacuum. Retest the system for normal operation

G3: INSPECT SYSTEM

	1 Switch the engine off
	2 Reconnect the vacuum hose
	3 Inspect the brake booster, rubber grommet, and all vacuum plumbing for cracks, holes, damaged connections, or missing clamps
	4 Pump the brake pedal several times to exhaust the vacuum. Push down on the brake pedal and hold
	Does the brake pedal move down when the engine is started? Yes Vacuum system is OK No GO to G4.

G4: CHECK POWER BRAKE BOOSTER VALVE

	1 Check the brake booster valve. For additional information, refer to Brake Booster component test in this section
	Is the power brake booster valve OK? Yes Check the brake booster. For additional information, refer to Brake Booster component test in this section. Install a new brake booster as required. Retest the system for normal operation No Install a new brake booster valve. Retest the system for normal operation

PINPOINT TEST H : BRAKE LOCKUP DURING LIGHT BRAKE PEDAL FORCE

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
H1: TEST BRAKE LOCKUP	
	1 Road test the vehicle and apply the brake pedal lightly
	Do the brakes lockup?

	Yes GO to H2. No No action required, vehicle is OK
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H2: INSPECT BRAKE PADS

	1 Inspect brake pads for contamination, correct installation, damage and type
	Has a concern been identified? Yes Correctly install or install new brake pads as required. Retest the system for normal operation No GO to H3.

H3: INSPECT BRAKE CALIPERS

	1 Inspect brake calipers for binding, leaking or sticking
	Has a concern been identified? Yes Correctly install or install new brake calipers as required. Retest the system for normal operation No No action required, vehicle is OK

PINPOINT TEST I : BRAKES DRAG

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
I1: ROAD TEST VEHICLE	
	1 Road test the vehicle and apply the brakes
	Are the brakes functioning correctly? Yes No action required, vehicle is OK No GO to I2.
I2: CHECK BRAKE CALIPERS	
	1 Check the front and rear calipers pistons and pins for binding, leaking or sticking
	Do the disc brake caliper pistons and pins bind, leak or stick? Yes Inspect the brake calipers and park brake cables. Install new components as required. Retest the system for normal operation No GO to I3.

I3: CHECK BRAKE BOOSTER

	1 Check the brake booster connecting rod alignment and travel
	Is the connecting rod OK? Yes Vehicle is OK No Install a new brake booster as required. Retest the system for normal operation

PINPOINT TEST J : EXCESSIVE/ERRATIC BRAKE PEDAL TRAVEL

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
J1: TEST ON ROUGH ROAD	
	1 Road test the vehicle on rough road conditions
	2 Apply the brakes slowly
	Is the brake pedal effort and brake pedal travel normal? Yes No action required, vehicle is OK No GO to J2.
J2: CHECK BRAKE FLUID LEVEL	
	1 Check the brake master cylinder reservoir fluid level
	Is the fluid level OK? Yes GO to J3. No Check brake master cylinder reservoir sealing points. For additional information, refer to Brake master cylinder component test in this section. Add brake fluid and bleed the brake system. Retest the system for normal operation
J3: CHECK BRAKE PEDAL RESERVE	
	1 Run engine at idle speed
	2 Apply the brake pedal lightly three or four times
	3 Wait 15 seconds for the vacuum to replenish
	4 Push down on the brake pedal until it stops moving downward or an increased resistance to the brake pedal travel occurs
	5 Hold the brake pedal in the applied position while increasing the engine speed to 2000 revs/min

	6 Release the accelerator pedal
	Does the brake pedal move downward as the engine speed returns to idle? Yes GO to J4. No Check the vacuum to the brake booster

J4: CHECK THE FRONT WHEEL BEARING ASSEMBLY

	1 Check the front wheel bearing assembly
	Are the front wheel bearings loose/damaged? Yes Tighten to specification or install a new front wheel bearing as required. Retest the system for normal operation No Check the front brake discs for thickness variances

PINPOINT TEST K : SLOW OR INCOMPLETE BRAKE PEDAL RETURN

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
K1: CHECK FOR BRAKE PEDAL RETURN	
	1 Run the engine at idle while making several brake applications
	2 Pull the brake pedal rearward with approximately 44.5 N (10lb) force
	3 Release the brake pedal and measure the distance to the toe board
	4 Make a hard brake application
	5 Release the brake pedal and measure the brake pedal to toe board distance. The brake pedal should return to its original position
	Does the brake pedal return to its original position? Yes No action required, vehicle is OK No GO to K2.

K2: CHECK FOR BRAKE PEDAL BINDING

	1 Disconnect the brake booster from the brake pedal. Check the brake pedal to ensure free operation
	Is the brake pedal operating freely? Yes Install a new brake booster as required. Retest the system for normal operation No

COMPONENT TESTS

BRAKE BOOSTER

1. Check all hoses and connections. All unused vacuum connectors should be capped. Hoses and their connections should be correctly secured and in good condition with no holes and no collapsed areas. Inspect the valve on the brake booster for damage
2. Check the hydraulic brake system for leaks or low fluid
3. With the automatic transmission in PARK, stop the engine and apply the park brake. Pump the brake pedal several times to exhaust all vacuum in the system. With the engine switched off and all vacuum in the system exhausted, apply the brake pedal and hold it down. Start the engine. If the vacuum system is operating, the brake pedal will tend to move downward under constant foot pressure. If no motion is felt, the vacuum booster system is not functioning
4. Remove the vacuum hose from the brake booster. Manifold vacuum should be available at the brake booster end of the hose with the engine at idle speed and the automatic transmission in PARK. Make sure that all unused vacuum outlets are correctly capped, hose connectors are correctly secured and vacuum hoses are in good condition. When it is established that manifold vacuum is available to the brake booster, connect the vacuum hose to the brake booster and repeat Step 3. If no downward movement of the brake pedal is felt, install a new brake booster
5. Operate the engine for a minimum of 10 seconds at a fast idle. Stop the engine and allow the vehicle to stand for 10 minutes. Then, apply the brake pedal with approximately 89 N (20lb) of force. The pedal feel (brake application) should be the same as that noted with the engine running. If the brake pedal feels hard (no power assist), install a

new valve and then repeat the test. If the brake pedal still feels hard, install a new brake booster. If the brake pedal movement feels spongy, bleed the brake system.

BRAKE MASTER CYLINDER

Usually, the first and strongest indicator of anything wrong in the brake system is a feeling through the brake pedal. In diagnosing the condition of the brake master cylinder, check pedal feel as evidence of a brake concern. Check for brake warning lamp illumination and the brake fluid level in the brake master cylinder reservoir

Normal Conditions

The following conditions are considered normal and are not indications that the brake master cylinder is in need of repair

- Modern brake systems are designed to produce a pedal effort that is not as hard as in the past. Complaints of light pedal efforts should be compared to the pedal efforts of another vehicle of the same model and year
- The fluid level will fall with brake pad wear

Abnormal Conditions

Changes in the brake pedal feel or brake pedal travel are indicators that something could be wrong in the brake system. The diagnostic procedure and techniques use brake pedal feel, warning indicator illumination and low brake fluid level as indicators to diagnosing brake system concerns. The following conditions are considered abnormal and indicate that the brake master cylinder is in need of repair:

NOTE:

Prior to carrying out any diagnosis, make sure the brake system warning indicator is functional

- Brake pedal goes down fast. This could be caused by an external or internal leak
- Brake pedal goes down slowly. This could be caused by an internal or external leak
- Brake pedal is low or feels spongy. This condition may be caused by no fluid in the brake master cylinder, reservoir cap vent holes clogged or air in the hydraulic system
- Brake pedal effort is excessive. This may be caused by a bind or obstruction in the pedal/linkage, a faulty non-return valve, booster or insufficient booster vacuum
- Rear brakes lock up during light pedal force. This may be caused by damaged brake pads, a partially applied park brake, a damaged ABS sensor or bearing failure
- Brake pedal effort erratic. This condition could be caused by the brake booster or incorrectly installed brake pads
- Brake warning indicator is on. This may be caused by low fluid level or float assembly damaged

NON PRESSURE LEAKS

Any reduced fluid volume in the brake master cylinder reservoir may be caused by two types of none pressure external leaks

Type 1: An external leak may occur at the brake master cylinder reservoir cap because of incorrect positioning of the gasket and cap. Reposition cap and gasket

Type 2: An external leak may occur at the brake master cylinder reservoir mounting seals. Repair such a leak by installing new seals and make sure that the brake master cylinder reservoir retaining bolt is correctly installed

DTC INDEX

For a list of diagnostic trouble codes (DTCs) that could be logged on this

vehicle, please refer to Section 100-00. REFER to: (100-00 General Information)

[Diagnostic Trouble Code Index: ABS Control Module - DTC: Anti-Lock Brake System Control Module \(ABS\)](#) (Description and Operation),

[Diagnostic Trouble Code Index - DTC: Electric Park Brake Control Module \(EPBCM\)](#) (Description and Operation).